

Late Deliveries & What Customers Say

Your own project: no other candidate has this one. Please do not share it.

Deadline	Friday 9 October 2026, 9:00 AM IST (72 hours)
Effort	About 6–8 hours of focused work, plus 10–15 min to get the data. A simple solution that works beats a big one that is unfinished
Data	A public dataset: links and steps in section 1 below. Sign-in needed: Free Kaggle account
Your track	Named in your project email. It decides your optional extra (section 6)
Submit	Reply to the project email with your GitHub link or zip, README, video link and 5 slides
Questions	Any time: reply to the project email to reach the Hiring Desk

The situation

Who is asking: The Head of Operations at an online marketplace.

The marketplace suspects late deliveries are hurting reviews, but nobody knows how often orders are late or what customers actually complain about.

Data: Kaggle — "Brazilian E-Commerce Public Dataset by Olist" (~1 lakh orders, 2016–2018). Use 3 of the 9 files: orders, order_items, order_reviews.

1. Get the data (about 10–15 min)

- Kaggle: Brazilian E-Commerce Public Dataset by Olist: <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

Steps

- Sign in to Kaggle (free) and click **Download**. The zip has nine CSV files.
- Use only these three: **olist_orders_dataset.csv**, **olist_order_items_dataset.csv** and **olist_order_reviews_dataset.csv**.

Check it loads (Python; run in Jupyter or Google Colab):

```
# pip install duckdb
import duckdb
for f in ['olist_orders_dataset.csv', 'olist_order_items_dataset.csv',
'olist_order_reviews_dataset.csv']:
    print(f, duckdb.sql(f"SELECT COUNT(*) FROM read_csv_auto('{f}').fetchone())
```

Keep the downloaded files out of your GitHub repository (add them to .gitignore), and write in your README exactly where they came from.

2. Part A: Data engineering (about 2–3 hours)

- Load the three files.
- Build one table with one row per delivered order: purchase date, promised (estimated) date, delivered date, days late (0 if on time), number of items, order value, review score.
- Write down how you avoided counting an order twice when joining items and reviews.
- Run 3 checks: order ID is unique in your table; delivered date is after purchase date; order value equals the sum of its items.

3. Part B: Analytics (about 2 hours)

- Share of orders delivered late.
- Late share by month (chart).
- Average review score for late vs on-time orders.
- Average score for orders more than 7 days late.

4. Part C: AI (about 1.5–2 hours)

- Take 200 reviews that have a comment. Use an LLM to tag each with one theme (late or not received, wrong or missing item, product quality, good experience, other) and write a one-line English summary.
- Check 30 by hand (use a translator) and report how many themes were right.
- Compare the theme mix for late vs on-time orders.

5. Your recommendation

Your recommendation slide must answer

What are the top three actions the Head of Operations should take?

6. Optional extra (only after Parts A–C work)

- **Which one:** only the row for **your track**, as named in your project email. Do not do all three.
- **When:** only after Parts A, B and C each work and your README is written. If you are short of time, skip it; it never makes up for a missing part.
- **Where to show it:** under "**9. Optional extra**" in your README, and in one line on your Results slide.
- **What it is worth:** up to 5 bonus points. It also gives us something to talk about in the interview.

Track	Optional extra
Data Platform	add the sellers file and compute the late share per seller (minimum 50 orders)
Analytics	late share by customer state
AI Applications	have the LLM extract the specific complaint and group them into five clusters

7. How to approach it

- **All three parts are required.** Each must work at a basic level. A simple, correct solution in every part beats an advanced solution in one part.
- **Tools:** any. Suggested: Python with pandas or DuckDB (SQL) in Jupyter or Google Colab. Do the data cleaning in code (SQL or Python) so it can be re-run. Charts can be in Python, Power BI Desktop or Excel.
- **AI tools:** use any LLM: a free API tier, a local model, or a free chat app used in batches. You may also use AI assistants to help write code. Say exactly how you used them in your README, and be ready to explain and change every line in the interview.
- **Data:** use only the dataset named in section 1. Do not add other companies' data.
- **Your own work:** discussing general ideas is fine; code, slides and video must be yours.
- **Stuck?** Ask. How you ask questions is part of how we work with customers. If something outside your control blocks you (power, internet, illness), tell the Hiring Desk before the deadline.

8. What to submit

Item	Requirement
Code	GitHub repository (public, or shared with the Hiring Desk) or a zip file. It must run from your README instructions

Item	Requirement
README	Use these headings: 1. The problem in your own words 2. Assumptions 3. Data source and how to run (where you downloaded the data, packages, order to run) 4. Part A: cleaning rules, with the number of rows affected by each, and your checks 5. Part B: key numbers and charts, each explained in one sentence 6. Part C: model or tool used, your prompt, how you checked it, and how often it was right 7. Limitations and what you would do next 8. How you used AI tools 9. Optional extra (if you did it)
Slides (5)	1) The problem · 2) Your approach · 3) Results (numbers and charts) · 4) Recommendation: answer the question above · 5) What you learned
Video	3–5 minutes (Loom, Google Drive or unlisted YouTube): walk through your slides and show your solution working

9. Free ways to use an LLM for Part C

Option	How
Google AI Studio (Gemini API)	Free API key; call it from Python
Groq	Free API key; fast open models
Ollama	Runs a small model on your laptop; no key needed, slower on older laptops
A chat app (ChatGPT, Claude, Gemini)	Paste 20–25 items at a time with the same prompt and ask for one CSV line per item

Free limits change often, so check them before you start. Give the model a fixed list of labels, ask for output you can parse, keep the same prompt for every batch, and save the prompt and raw outputs in your repository.

10. How your work is assessed

Area	Points	What we look for
Understanding and assumptions	10	You restate the problem correctly and list sensible assumptions
A. Data engineering	20	Cleaning rules written down with row counts; the right grain; checks run
B. Analytics	20	Correct, reconciled numbers; readable charts; each number explained in plain words
C. AI with an honest check	15	A sensible prompt; results checked against the truth; accuracy reported honestly, with failures shown
Communication	20	README, slides and video a non-technical manager can follow; a recommendation backed by numbers
Judgement and honesty	15	You notice data problems, state limits, and invent nothing

We are generous about polish and strict about honesty. "The LLM got 22 of 30 right, and here are the 8 it missed" scores better than a claim that looks impressive but does not match the data.

11. What happens next

When	What
Friday 9 October, 9:00 AM IST	Deadline. The Hiring Desk confirms it has received your submission
By Sunday 11 October, evening	The Hiring Desk writes to every candidate about the next step. Good luck, and enjoy the problem

When	What
12–16 October	A 75-minute one-to-one interview for shortlisted candidates, on video: • 5 min: getting to know you • 10 min: you present your 5 slides • 15 min: you walk through your code and data choices • 15 min: you make a small change to your own code, live • 15 min: a conversation about your projects, what you have learned, and how you work with others • 10 min: your questions for us • 5 min: next steps